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`https://github.com/julianmak/OCES5303\_ML\_ocean`

The repository principally contains the compiled products rather than the source for size reasons.

- ▶ Associated Python code (as Jupyter notebooks mostly) will be held on the same repository. The source data however might be big, so I am going to be naughty and possibly just refer you to where you might get the data if that is the case (e.g. JRA-55 data). I know I should make properly reproducible binders etc., but I didn't...
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OCES 5303 : ML methods in Ocean Sciences

Session 2: more `sklearn`, cross-validation

Outline

- ▶ more sklearn things
 - linear models such as Linear Regression, Ridge, Lasso, Elastic-Net (cf. exercise in varying the loss function)
 - idea behind gradient descent
 - dimension reduction (e.g. PCA, lle, *t*-SNE)
 - clustering (e.g. *K*-means, others)
- ▶ cross validation and model hyper-parameter tuning
 - *k*-folds cross validation
 - ensemble methods

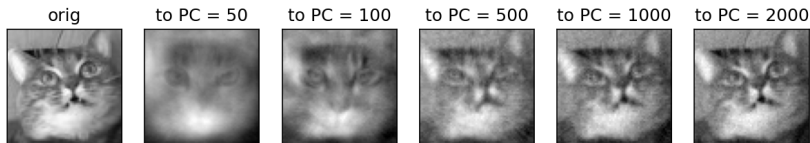


Figure: Result from an eigencat projection (it's really just PCAs).

Linear models

- ▶ given multiple features $X = (X^{(1)}, X^{(2)}, \dots)$ and a **single** target Y , a **linear** model considers

$$\hat{Y} = a_0 + a_1 X^{(1)} + a_2 X^{(2)} + \dots a_N X^{(N)}$$

→ the model parameters or control variables are $\{a_0, a_1, \dots, a_N\}$

→ basically a linear algebra type problem (e.g. invert the matrices)

→ different methodologies take different loss functions J (see later)

- ▶ note: this is not multi-**variate** regression where we have **multiple** targets Y , but see **GLMs**

Linear models: Linear Regression

- ▶ standard linear regression takes

$$J = \|\hat{Y} - Y\|_{L^2}^2 = \sum_{i=0}^M |\hat{Y}_i - Y_i|^2,$$

where Y_i is the training data and $\hat{Y}_i$ are the predictions from the model

→ note lower index, M samples, but N lots of features described by $X^{(j)}$

→ this basically provides the 1d LOBF embedded in $(N + 1)$ d space

→ can have issues when features (the $X^{(j)}$) are not independent of each other

→ can be sensitive to data noise

Linear models: Ridge

- ▶ consider controlling the size of the model coefficients a by

$$J = \|\hat{Y} - Y\|_{L^2}^2 + \alpha \|a\|_{L^2}^2$$

→ this is **penalisation** on a with strength α

- ▶ extra **hyper-parameter** α here
→ large α means small coefficients and control colinearity,
but at then expense of skill
- ▶ note: different α 's needed if data is not standardised

Linear models: Lasso

- ▶ could even limit number of features by asking for as many zero a_j as possible, via

$$J = \frac{1}{2M} \|\hat{Y} - Y\|_{L^2}^2 + \alpha \|a\|_{L^1}$$

→ large α we want to keep as many non-zero coefficients as possible, but at then expense of skill

→ related to **compressed sensing**

Linear models: Elastic-nets

- could do a bit of both via

$$J = \frac{1}{2M} \|\hat{Y} - Y\|_{L^2}^2 + \alpha \rho \|a\|_{L^1} + \frac{\alpha(1-\rho)}{2} \|a\|_{L^2}^2$$

→ ρ is a ratio hyper-parameter controlling sparsity and model parameter values

Sample loss function + gradient descent

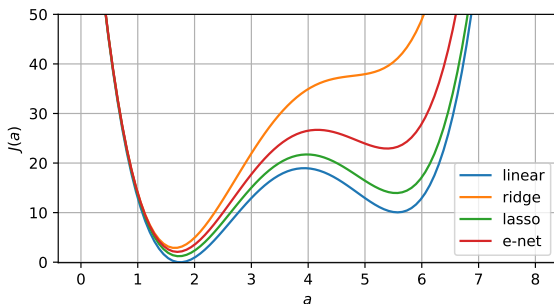


Figure: Made up 1d example of $J = J(a)$ with different choices of regularisation,
 $J = (1/2M)\|\hat{Y} - Y\|_{L^2}^2 + \alpha\rho\|a\|_{L^1} + (1/2)\alpha(1 - \rho)\|a\|_{L^2}^2$.

- optimise by finding minimum, extremum occurs where the derivatives are zero $\Rightarrow$ root finding problem
→ **gradient** information is used to find direction of **descent**

Example: regressing to polynomial data

- ▶ going to do a demonstration case of taking

$$Y = X^2 + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma)$$

where σ controls the level of ‘noise’, but going to fit this to

$$\hat{Y} = a_0 + a_1 X^1 + a_2 X^2 + \dots a_{10} X^{10}$$

i.e. my features $X^{(j)}$ in this case happens to polynomials X^j
(X raised to power j)

→ might expect to get $a_2 = 1$ and small values elsewhere

→ a slightly more subtle meaning to “linear” is needed...

Example: regressing to polynomial data

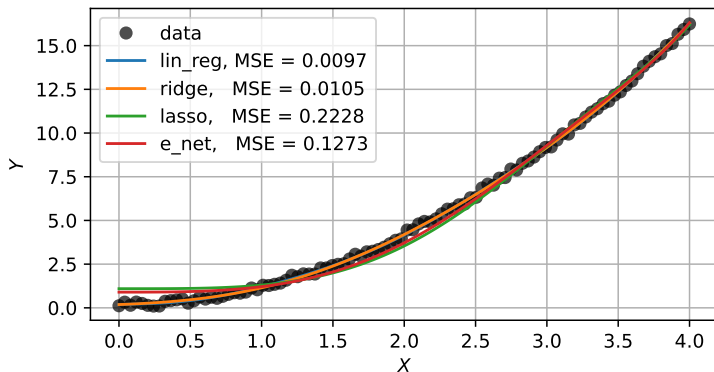


Figure: Polynomial regression for some contrived data, with MSE scores printed on.

Example: regressing to polynomial data

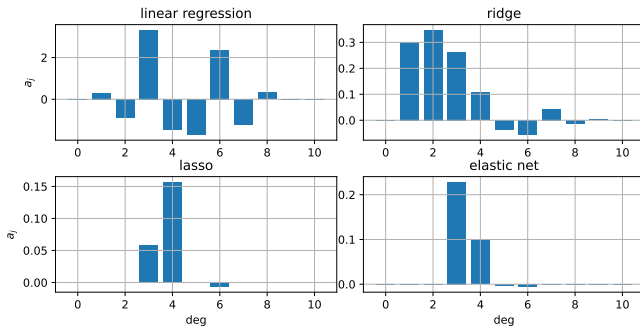


Figure: Polynomial regression for some contrived data, with MSE scores printed on.

- ▶ notice a_2 is not always largest!
- ▶ notice L^1 based penalisation (lasso and elastic net) gives more small coefficients
→ fewer features (more likely not over-fitted?)

Example: image classification + perceptrons (see Lec 04)

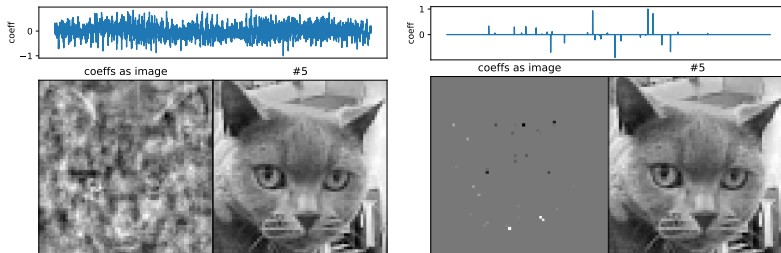


Figure: Visualising a perceptron from a pseudo-inversion of a matrix. (Left) Standard linear regression (L^2 inversion). (Right) Lasso (L^2 with a L^1 penalisation). See Lec 04.

- **neural networks** in its simplest form is just matrix inversion (if the **activation function** is trivial)
 - can obtain the model via a (pseudo-)inversion
 - basically an optimisation problem, can design loss functions accordingly

Dimension reduction: PCAs

- Q. if not all features are as important, can we find the important ones *a priori*?
→ would help presumably against over-fitting

Dimension reduction: PCAs

- Q. if not all features are as important, can we find the important ones *a priori*?
→ would help presumably against over-fitting
- ▶ **Principal Component Analysis** (PCA) is one way
→ shows up also as **Empirical Orthogonal Functions** (EOFs) particularly when spatial patterns are involved
 - ▶ linear combinations of features that explain the most variance in the data
→ effectively does a **SVD**, returns
 1. singular values σ (related to variance explained)
 2. the left and right singular vectors that gives the new co-ordinate system and relevant expansions

Dimension reduction: PCAs

- easier to explain visually with the penguins data
→ data has been standardised here

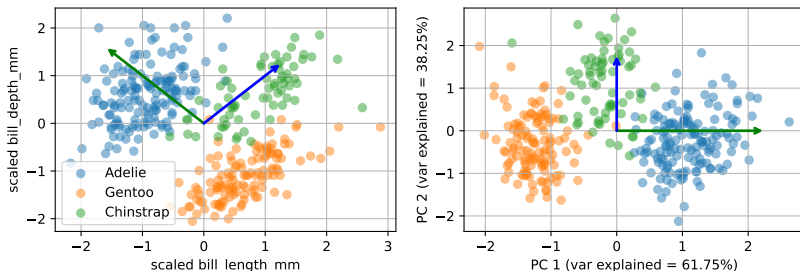


Figure: (Left) Original (but standardised) data with two choices of variables. (Right) The same but in the principal component representation.

- for this case it really is just a co-ordinate transformation

Dimension reduction: PCAs

- below shows dimension reduction (from 4 to 2)

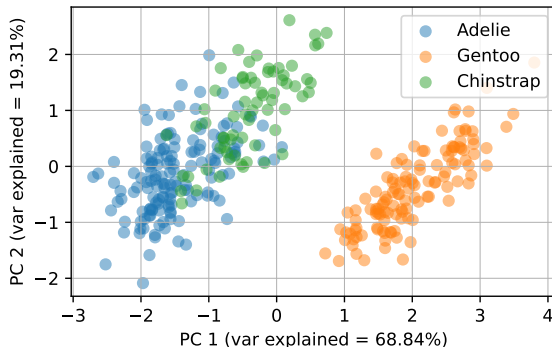


Figure: Representation in PC1 and PC2 co-ordinate system.

- note the variance explained does not sum to 100% here
→ this is now a **projection** with information loss

Dimension reduction: manifold methods

(this part makes more sense in about 15 mins)

► other manifold based methods

→ Locally Linear Embedding (preserves local distances, think local PCAs to give global co-ord)

→ *t*-distributed Stochastic Neighbor Embedding (*t*-SNE; based on pdfs, good at picking out local structures)

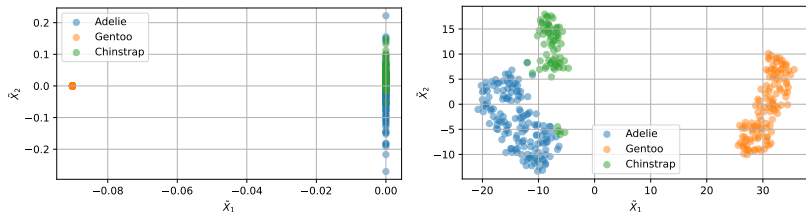


Figure: Demonstration of Locally Linear Embedding and *t*-SNE on the penguins data.

Demonstration: eigencat

- ▶ example demonstrating PCA on images of cats
→ feature extraction, cf. facial recognition
- ▶ message data into (n_cat, pixels)
then throw into PCA

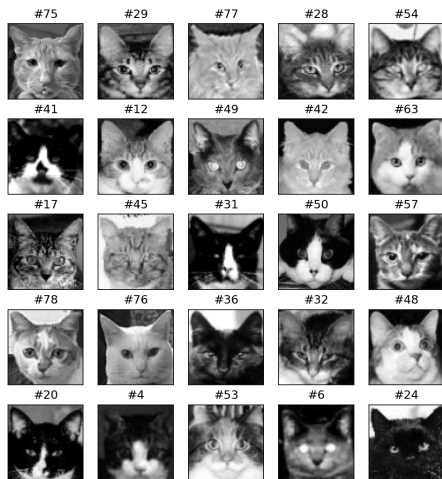


Figure: Data in `cats.csv`.

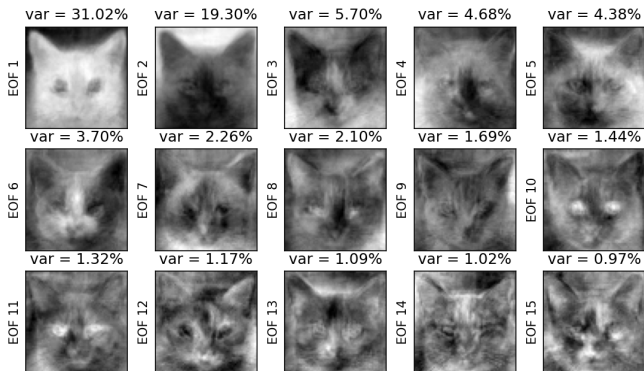
Demonstration: eigencat

► want

$$\text{Image}(\text{pixels}) = \sum_i \text{PC}_i \times \text{EOF}_i(\text{pixels}),$$

→ call the patterns the EOFs, PCs are the loadings

→ reshape EOFs to get "averaged" cat faces



Demonstration: eigencat

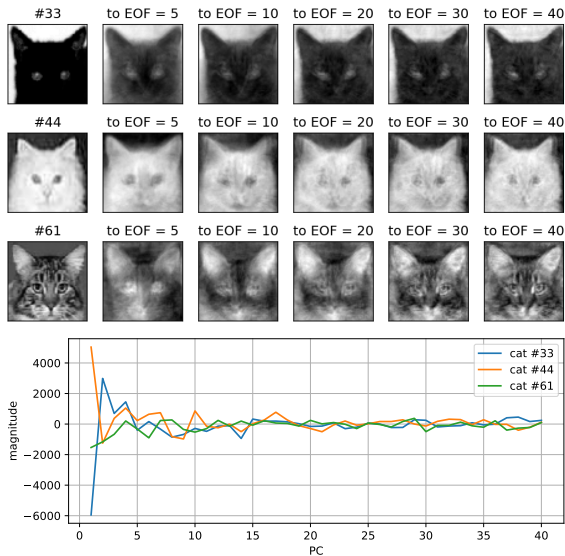


Figure: Expansion and loadings of three of the samples.

Demonstration: eigencat

- ▶ different cats will have a different PC signatures, use that for
 - classification?
 - features for model training? (much lower dimension)
- ▶ warning: can start doing ridiculous things like below if my data space is large enough!
 - next few pages show the same thing if I use an expanded dataset (2000 images)

Demonstration: eigencat

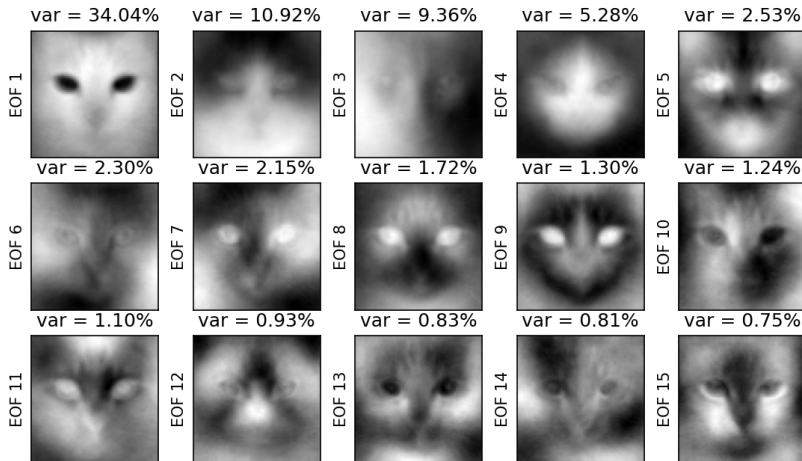


Figure: Eigencats from `cats_bw_enlarged.csv`.

Demonstration: eigencat

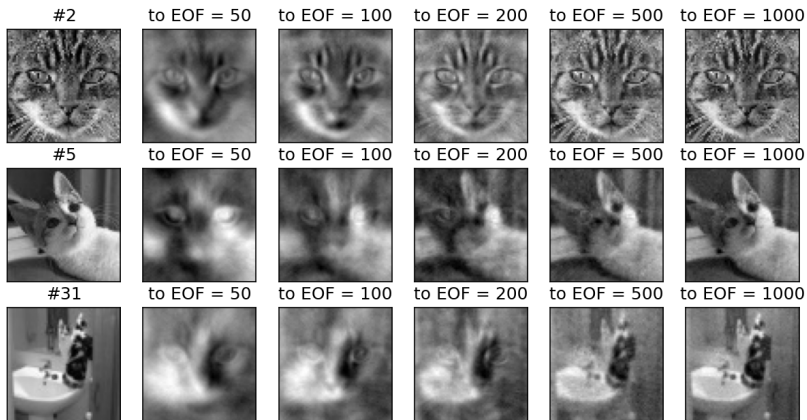


Figure: Expansion of a few selected images from the expanded dataset.

Demonstration: eigencat

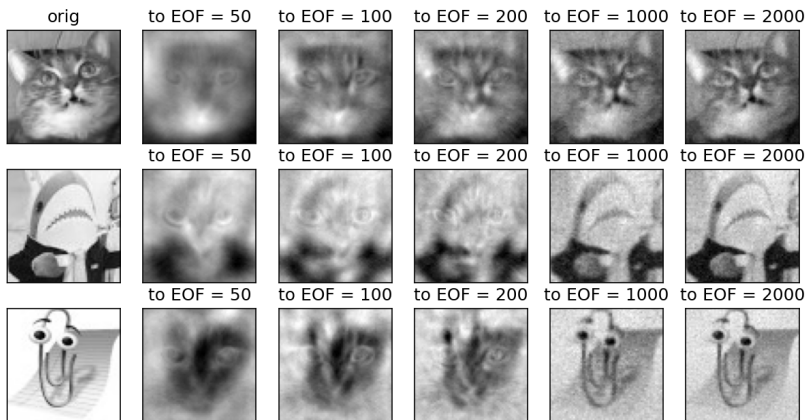


Figure: Eigencat expansion of the three ad hoc TAs (which were not in the dataset).

Clustering: a motivation

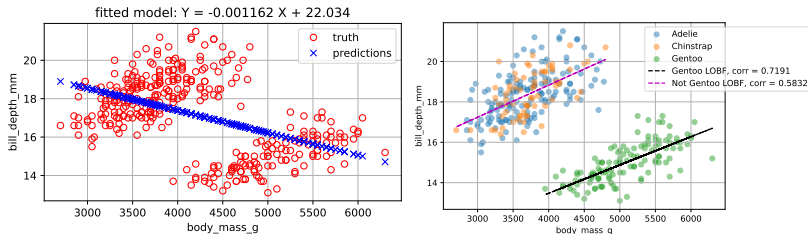


Figure: Contrived example of completely different models depending on data selection.

- if we had no labels in penguins data the linear regression is pretty bad (although it was never going to be good...)
 - use **clustering** to create labels?
 - e.g. use that to inform training of model, or to train different models

Oceanographic examples

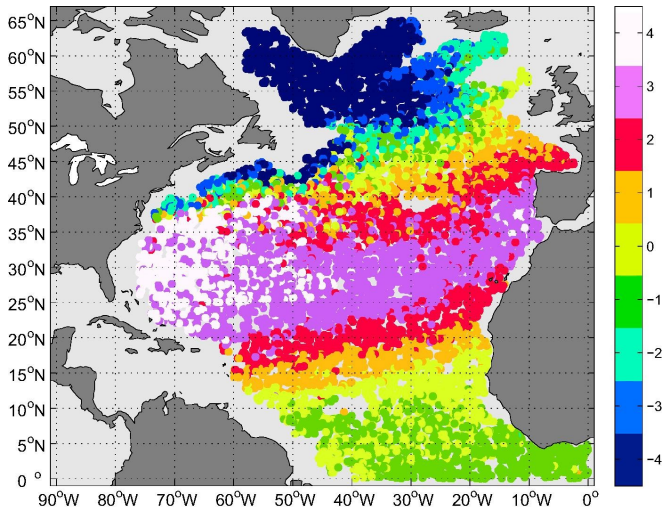


Figure: From Maze *et al.* (2017), Fig. 4. Gaussian Mixture Model to identify watermass clusters from Argo data in Atlantic.

Oceanographic examples

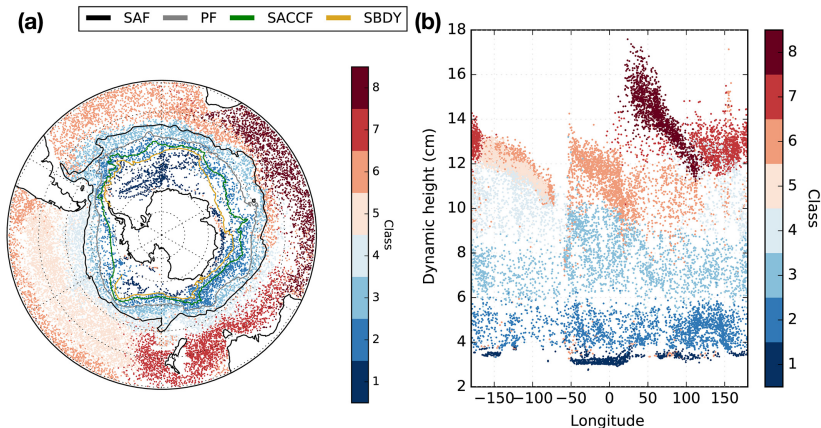


Figure: From Jones *et al.* (2019), Fig. 5. Gaussian Mixture Model to identify watermass clusters from Argo data in Southern Ocean.

Oceanographic examples

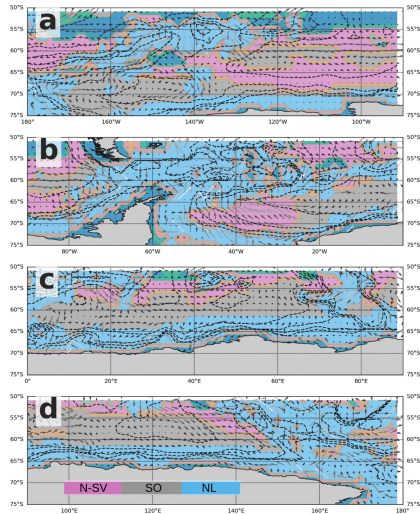


Figure: From Sonnewald *et al.* (2023), Fig. 4. *k*-means to identify clusters based on dynamic (from barotropic vorticity budget).

Oceanographic examples

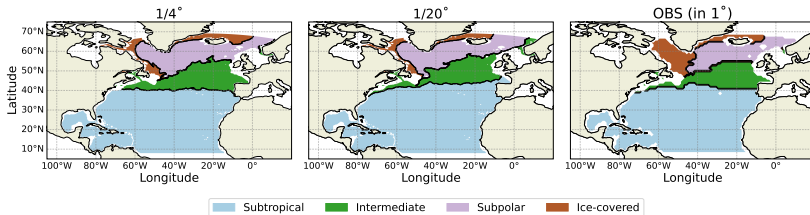
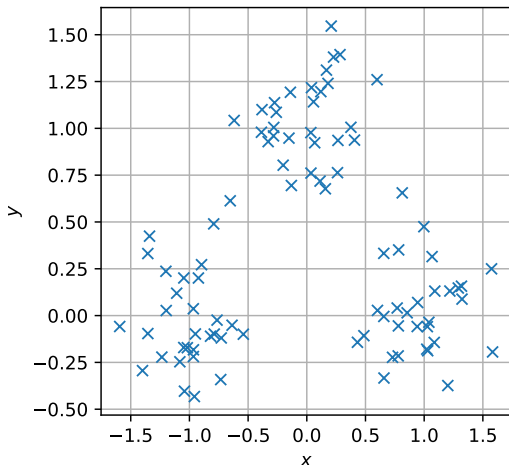


Figure: From Ruan *et al.* (in prep.), which identifies regions depending on biogeochemical activity.

- hierarchy building of clusters using various variables
 - SST, MLD, sea-ice fraction, not (lon, lat)
 - nitrate concentration, surface chlorophyll concentration
- aim to classify POC and DOC export, but do not include those in building of cluster

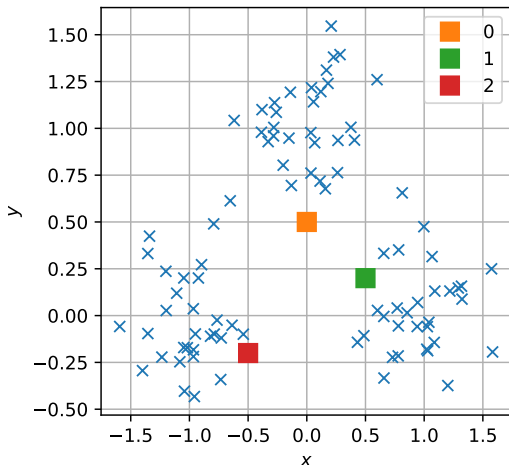
e.g. K -means

- demonstrate the algorithm with K -means
 - artificially create some data ($K = 3$ is sensible)



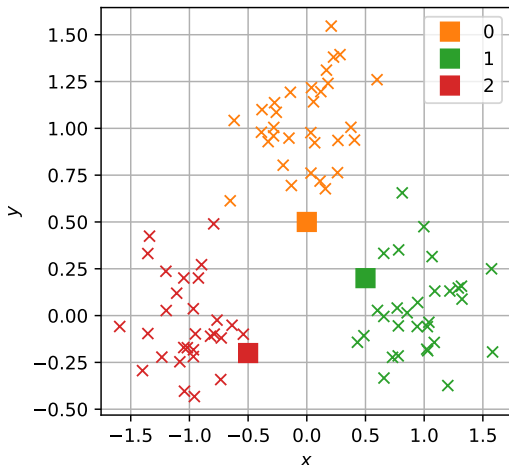
e.g. K -means

- provide initial centres of clusters
→ K -means is quite robust, can just guess



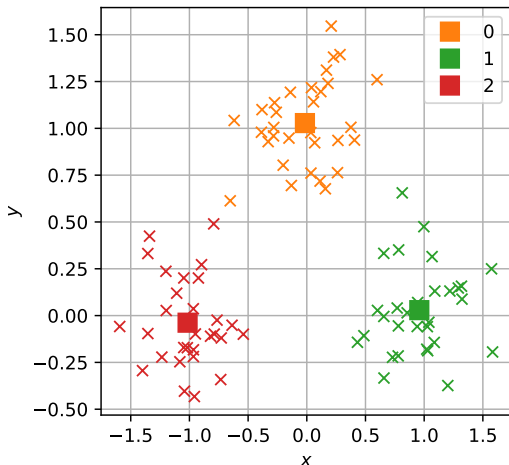
e.g. K -means

- label points closest to centres accordingly
→ distance dependent, normally L^2 (Euclidean distance)



e.g. K -means

- from new cluster, find the new location of the centre
→ update and iterate accordingly



K-means in sklearn

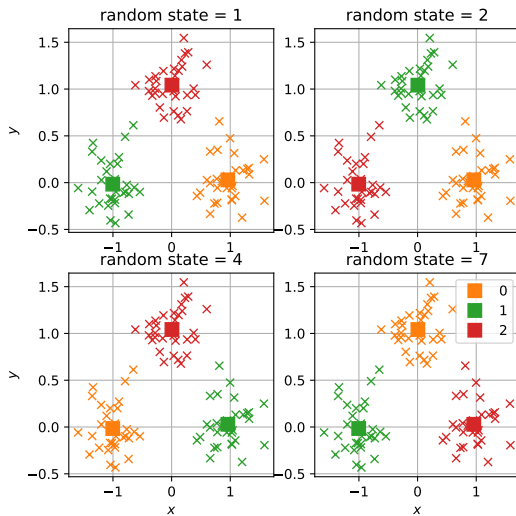


Figure: K-means through `sklearn`. Random state changes initial guess.

A case where K -means as is will never(?) work

- ▶ all of the above depends on the choice of 'distance' again

A case where K -means as is will never(?) work

- ▶ all of the above depends on the choice of 'distance' again
- ▶ there are well-known examples where the L^2 distance is simply not the relevant one
 - e.g. the moon data (2d example) and Swiss roll data (can do 2d or 3d)



Figure: Moon Moon and two rolled up towels resembling a Swiss roll.

A case where K -means as is will never(?) work

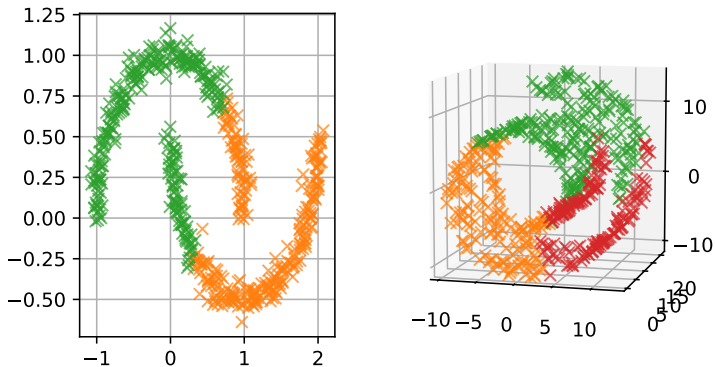


Figure: K -mean as applied to the crescent moons and swiss roll data.

The issue of 'distance'

- ▶ the data may have structure and lives on a curve/surface/volume etc. (e.g. manifold up to noise)
 - manifold may have a lower dimension structure (1d and 2d here)
 - but embedded in an ambient space ($\mathbb{R}^2$ and $\mathbb{R}^3$ here)

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- ▶ it is the distance **intrinsic** to the manifold that is presumably of interest
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- ▶ find a suitable embedding, as in dimension reduction previously?
 - reduce dimension first, then cluster

Moon data demonstration

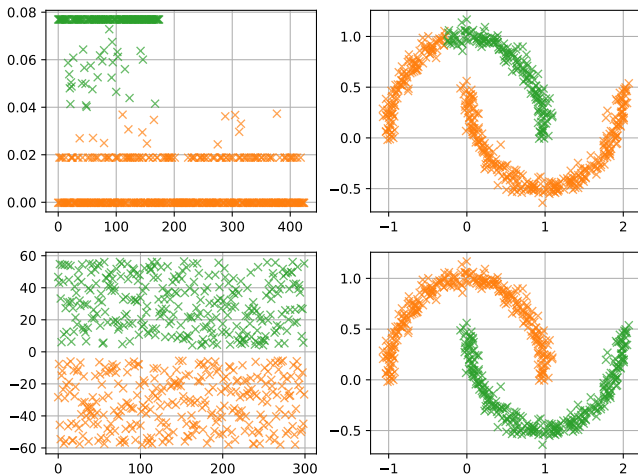


Figure: 1d projection (of 2d data) via (top) LLE and (bottom) *t*-SNE before *K*-means. LLE can work on occasions, but *t*-SNE seems more robust.

Swiss roll data demonstration

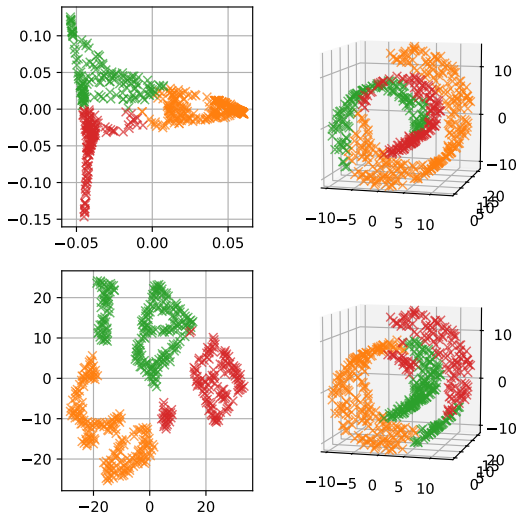


Figure: 2d projection (of 3d data) via (top) LLE and (bottom) *t*-SNE before *K*-means. LLE arguably segments better.

Other algorithms (e.g. DBSCAN)

- ▶ other algorithms do things differently
- ▶ DBSCAN considers clusters as high density regions and gaps are low density regions
 - two parameters: radius ϵ of some ball and number of points within said ball to quantify 'density'
 - number of clusters identified is a result of the above two choices (unlike in *K*-means)
 - can deal with non-Euclidean distances (you need to provide it though)
- ▶ hyper-parameter tuning + cross-validation needed!

Other algorithms (e.g. DBSCAN)

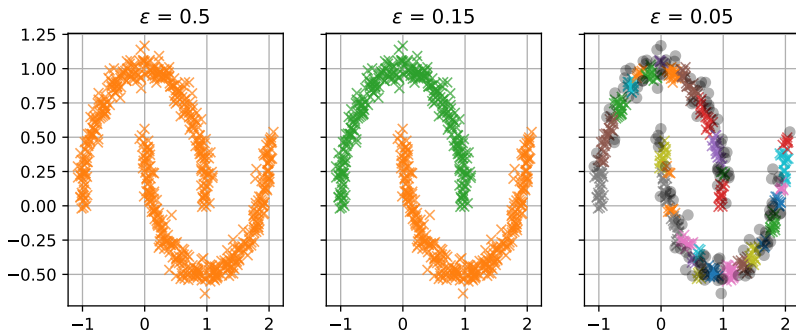


Figure: DBSCAN on moon data at varying ϵ .

- optimal ϵ for the two clusters
- if too small the too many clusters identified
→ black points are 'noise' points (no confidence in which cluster it should fall in)

Other algorithms (e.g. DBSCAN)

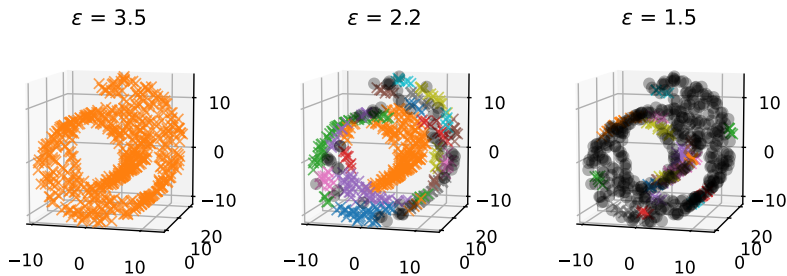


Figure: DBSCAN on swiss roll data at varying ϵ .

- ▶ optimal ϵ for the one giant cluster (bit contrived though...)
- ▶ smaller ϵ leads to segmenting along the surface, so ok

Demonstration: penguins data

- K -means ($K = 3$) on full 4d data (standardised per feature), then compare classification skill
 - some manual remapping needed
 - skill is around 91% accuracy for this realisation

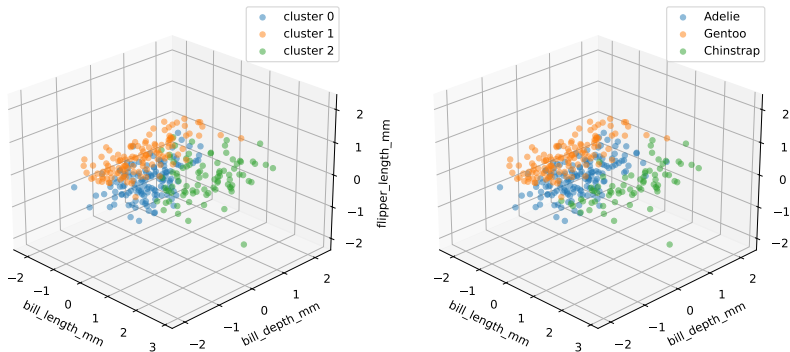


Figure: K-means on standardised penguins data.

Demonstration: penguins data

- ▶ as above but with a t -SNE to 2d before K -means
 - some manual remapping needed
 - skill is around 96% accuracy for this realisation

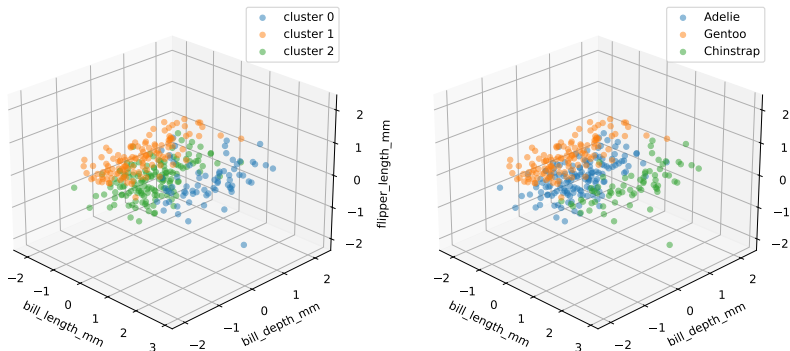


Figure: t -SNE to 2d before K -means on standardised penguins data.

Validation and model robustness

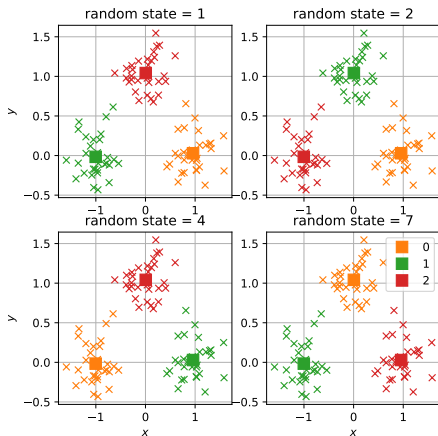


Figure: K-means through `sklearn`. Random state changes initial guess.

- ▶ there is inherent **randomness** in the training
 - e.g. from train-test split
 - resulting models can have some randomness also
 - how do we know our model 'works' not just because we got 'lucky'?

Validation and model robustness

- ▶ just do what we would normally do with statistics, e.g.
 - train an **ensemble** and take some averages
 - evaluate **sensitivity** to choices made
 - quantifying/probing for **uncertainties**
 - evaluate on possible **over-fitting**
 - ...

**what you wouldn't do with statistics
you should not do with machine learning**

Cross-validation

- ▶ **over-fitting** is when model has skill in training, but may suck in test case
 - particularly if testing data is outside the pdf of the training data
- ▶ e.g. high degree polynomial fitting

$$\hat{Y} = \sum_{i=0}^p a_i X^i = a_0 + a_1 X^1 + a_2 X^2 + \dots$$

- the degree of polynomial p could be regarded as the **model hyperparameter** here
 - blows up quickly outside of training range

- ▶ probably best to regard most ML models as over-fitted unless shown otherwise...

e.g., k -fold cross validation

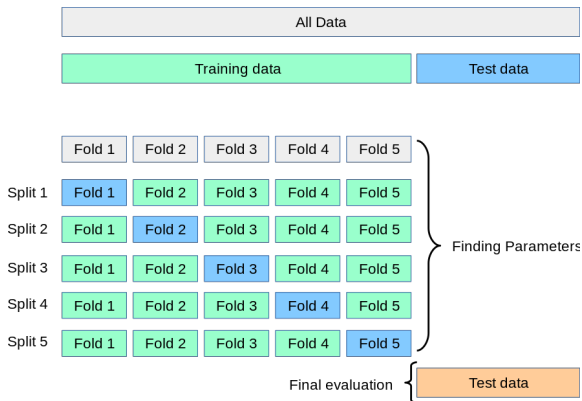


Figure: Schematic of k -fold cross-validation. Taken from https://scikit-learn.org/stable/modules/cross_validation.html.

- ▶ do repeated training on the folds, then do testing
→ the training thus also serve as **validation set**

Cross-validation

- ▶ then select 'best' model, e.g.
 - select best choice of p
 - select the $\{a_i\}$ that performs best during training
 - average the $(a_0, a_1, \dots)$ and use the “averaged” model
 - ...
- ▶ similarly if having a whole load of different models
 - for linear models next lecture, do the above but for linear regression, LASSO, ridge, etc.

Demonstration

It looks like you're trying to overfit a model.

Would you like help?

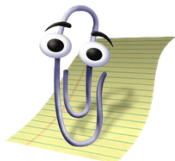


Figure: The OG AI (which was mega annoying by the way).

- ▶ (deterministic) linear models as variations of the loss function
- ▶ examples of dimensional reduction and feature identifications
- ▶ clustering as a way to create features
- ▶ need for hyper-parameter tuning and model validation

**what you wouldn't do with statistics
you should not do with machine learning**

Moving to a Jupyter notebook →